CodeLit

A Windows Desktop and Web Application for Secure Exam Lockdown and Automated Programming Assessment

in Philippine Computing Education

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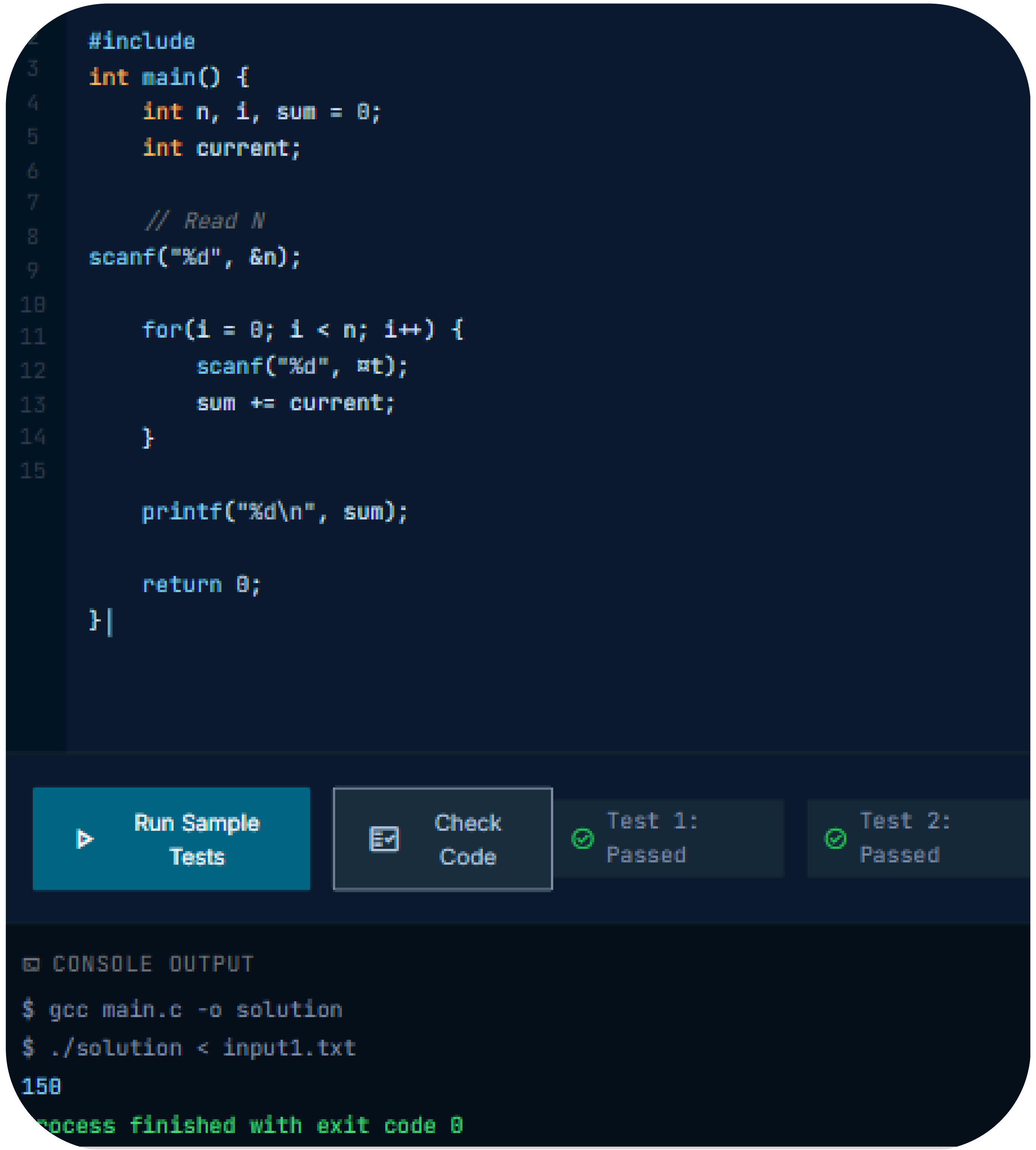


The Problem

No Desktop Lockdown

Cloud tools only restrict browsers — Alt+Tab, Task Manager, and OS-level shortcuts remain open.

Server-Side Bottlenecks

Simultaneous submissions to remote grading servers cause delays during time-limited exams.

Unreliable Connectivity

Many Philippine schools face unstable internet, making cloud-only platforms risky for live exams.

No Unified Platform

No single tool combines system-level security AND automated code assessment together.

Existing Solutions & Gaps

CodeChum

Strengths

- Automated code assessment
- Multi-language support (C, C++, Java...)
- Real-time performance monitoring
- SECUNA-certified, NPC-registered

Limitation

Browser-only, cannot enforce OS-level lockdown (Alt+Tab, Task Manager, Windows key).

Safe Exam Browser (SEB)

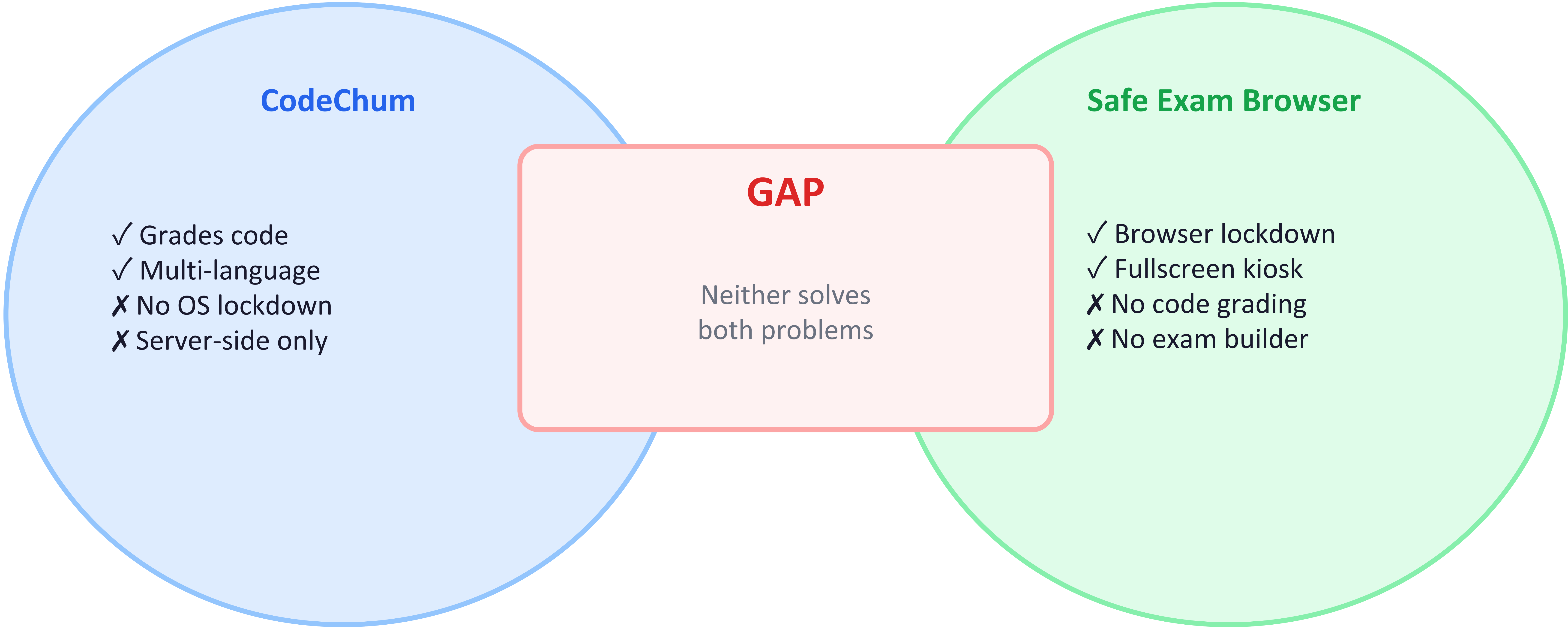
Strengths

- Enforces fullscreen, restricted browser
- Encrypted exam config files
- LMS integrations (Moodle, Canvas)
- Free and open-source

Limitation

No code evaluation, compilation, or programming assessment of any kind.

The Core Gap



No single platform currently provides secure OS-level lockdown + local code execution + automated assessment.

Proposed Solution

CodeLit

A Windows desktop application that combines system-wide exam lockdown with local automated programming assessment, supported by cloud-based exam management, synchronization, and monitoring.

System-Wide Lockdown

Intercepts Alt+Tab, Windows key, Task Manager, copy-paste, screenshots at OS level

Local Code Compiler

Compiles and executes C programs locally via GCC/MinGW — no server needed

Teacher Dashboard

Live monitoring, violation alerts, and session control from any browser

Multi-Type Exam Builder

Multiple Choice, Fill-in-the-Blank, and Coding questions in one exam

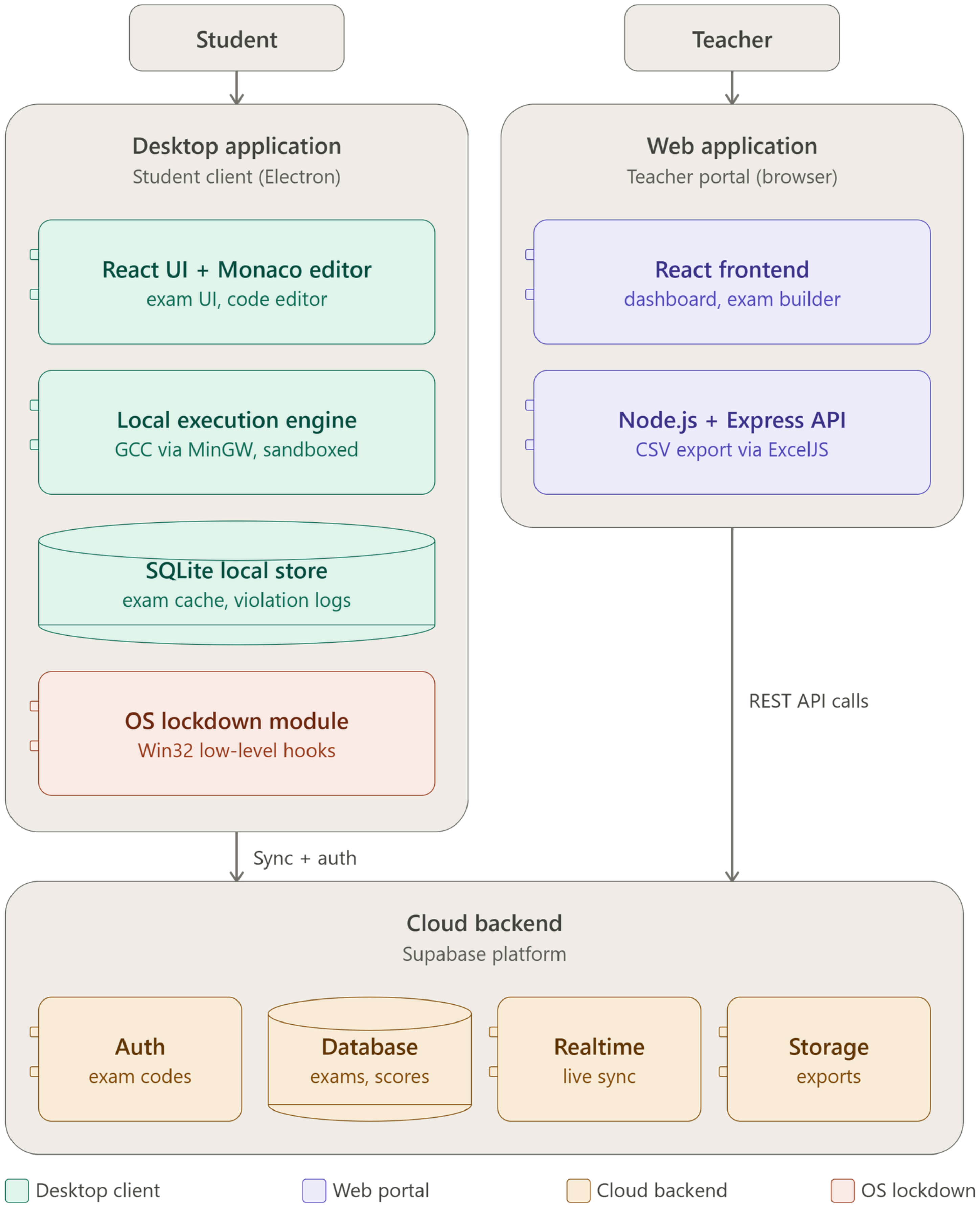
Cloud Sync & Storage

Centralized records, submissions, and violation logs stored securely

Export Reports

CSV result sheets and incident reports generated instantly after the exam

System Architecture



Two Entry Points

Student Side - Desktop App

- Monaco editor + local GCC/MinGW compiler (grades on-device)
- SQLite caches exam data locally
- OS Lockdown Module disables Alt+Tab, Start menu, copy-paste, Task Manager via Win32 hooks

Teacher Side - Web App

- React dashboard + exam builder
- Node.js/Express handles exam creation and CSV export

Cloud Backend

- Auth → single-use exam codes
- Database → exams, scores, violations
- Realtime → live teacher dashboard updates
- Storage → exported files

Key Design Principle

- Grading and lockdown happen locally, bandwidth doesn't affect exam performance
- The cloud's only job is keeping records accurate and the teacher's dashboard live

Lockdown & Security

What's Intercepted & Blocked

- Alt+Tab and all app-switching shortcuts
- Windows key and Start menu
- Task Manager (Ctrl+Shift+Esc, Ctrl+Alt+Del)
- Copy-paste (Ctrl+C / Ctrl+V / Ctrl+X)
- Screenshot tools (PrintScreen, Win+Shift+S)
- Disallowed applications (process monitoring)
- System audio (muted on launch)

How It Works

1. Teacher starts exam → lockdown activates simultaneously on all student machines
2. SetWindowsHookEx (WH_KEYBOARD_LL) intercepts OS-level keyboard input in Ring 3 userspace
3. Fullscreen always-on-top kiosk window captures focus
4. Active process monitoring flags disallowed apps instantly
5. Bypass attempts silently logged with timestamps on teacher dashboard
6. Lockdown released only when teacher explicitly ends the session

Assessment Engine

1. **Student Writes Code**
   Monaco Editor (VS Code engine)

2. **Checks Against Samples**
   Visible test cases before submit

3. **Final Submission**
   Point deduction per check attempt

4. **Local GCC Compiles**
   Process isolation, time limits

5. **Hidden Test Cases Run**
   Teacher-defined, never visible

6. **Score Returned**
   Partial credit + attempt deductions

How CodeLit Fills the Gap

Gap	CodeChum	SEB	CodeLit
OS-level Desktop Lockdown	✗	Partial	✓
Automated Code Grading	✓	✗	✓
Local Compilation (No Server)	✗	✗	✓
Coding-Specific Exam Builder	✓	✗	✓
Violation Logging & Reports	Partial	✗	✓
Works with Unstable Internet	✗	Partial	✓
Unified Exam Workflow	✗	✗	✓

CodeLit

Secure. Local. Integrated.

OS-Level Security | **Local Code Execution** | **Cloud Management**

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